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Development of degraded ravine lands of Western India using Sapota (*Achras zapota*) plantation with terracing vs. trenching-on-slope-based conservation measures

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Abstract

Globally, ravine landscapes are considered among the world's most degraded ecosystems. Restoring ravines is considered a high priority item in the conservation programmes, and tree establishment augmented with appropriate soil and water conservation measures is the most sought-after restoration strategy. This study examined the impact of terracing and trenching on runoff, soil loss, and soil properties along with Sapota (*Achras zapota*) growth, fruit yield, biomass and carbon stock during 6 years (2010–2015) in a degraded ravine land of Western India. The four soil and water conservation strategies tested were (1) bench terracing + Sapota (Bt), (2) bench terracing + Sapota with crop cultivation (BtCr), (3) continuous slope + Sapota (Sl), and (4) continuous slope + Sapota with trenches (SlTr). All conservation techniques increased Sapota height (Bt: 54%; BtCr: 27%; SlTr: 35%) and basal diameter (Bt: 36%; BtCr: 18%; StCr: 22%) compared to Sl. Biomass and carbon stock of Sapota were in the order of Bt > SlTr > BtCr > Sl. During the droughtperiod, fruit yield was highest in Bt followed by BtCr, SlTr, and least in Sl. The treatment SlTr and Bt reduced runoff by 16–34% and soil loss by 15–25%, compared to Sl. The growth and biomass were strongly correlated with soil loss. The findings suggested that bench terracing is the best soil and water conservation measure for restoring highly degraded ravines of Western India. Under circumstances where terracing is not feasible, trenching could be a significant step along with Sapota plantation to restore these ravines.

KEYWORDS

carbon stock, ravine, Sapota plantation, soil and water conservation, terracing, trenching

1 | INTRODUCTION

Over one billion ha of the world's land is affected by physical, chemical and biological degradation making it unproductive for agriculture (Cai, Zhang, Wang, 2011; Munoz-Rojas et al., 2012). Drylands, which include arid and semiarid environments, are prone to continuous land degradation with estimates suggesting an existence of around 10–20% of such ecosystems globally (Millennium Ecosystem Assessment, 2005; Safriel et al., 2005), and millions of hectares every year are getting degraded (Brauch & Spring, 2009; Wang, Xue, Zhou, & Guo, 2015; Yan & Cai, 2015). This problem is particularly serious in semiarid climates, where soil erosion leads to the development of

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highly degraded ravine landscapes (Rao, Gaur, Kumar, Kurothe & Tiwari, 2013). Morphologically, a ravine is a small, narrow, steep-sided valley that is larger than a gulley but smaller than a canyon and is formed by water erosion. Globally, a ravine landscape is considered as one of the most degraded ecosystems (Ali, Sethy, Singh, Parandiyal, & Kumar, 2016). Restoring these ecosystems is an important step toward conserving precious soil resources, maintaining hydrological balance, enhancing soil carbon (C) sequestration, mitigating climate change, and improving socio-economic conditions and livelihood security of people in the region (de MoraesSá et al., 2015; Minnemeyer, Laestadius, & Sizer, 2011; Perring et al., 2015; Prosdocimi et al., 2016; Roa-Fuentes, Martínez-Garza, Etchevers, & Campo, 2015).

Several types of anthropological activities such as improper management, illicit mining, uncontrolled grazing, deforestation, and land-use change (Keesstra, 2007; Navar & Synnott, 2000; Sikka, Mishra, Singh, Rao, & Islam, 2016), etc. accelerate the soil erosion processes which subsequently transform an area into a ravine landscape. The unchecked exploitation results in long-term changes in the soil environment (Lal, 2001, 2016), resulting in poor plant growth (Shrivastava & Kumar, 2015), and altered ecosystem functions (Evans, Zipper, Burger, Strahm, & Villamagna, 2013). However, soil and water conservation measures (SWCM) along with afforestation have good potential to improve and sustain the biological productivity of degraded ravine lands (Ali et al., 2016; Kurothe et al., 2014). Such Conservation measures can achieve good success if implemented with appropriate incentive mechanisms and subsidies to the local farmers' by the regional governments (Biratu & Asmamaw, 2016).

To restore unproductive ravines, soil and water conservation measures, for example, contour bund, peripheral bund, terracing, check dams, trenching and agroforestry (Chaturvedi, Kaushal, Tomar, Prandiyal, & Panwar, 2014; Kurothe, Vishwakarma, et al., 2014) have been successfully tested and region-specific recommendations have been developed. Of all the measures, reforestation has been considered as the most important tool to mitigate land degradation in dry environments (Hu, Niu, Zeng, & Wang, 2015). However, paucity of native commercial tree species for reforestation of ravine lands is a big constraint. The dry, degraded ravines of Western India are characterized by the dominance of low commercial value native trees (Parandiyal, Samara, Singh, Singh, & Rathore, 2000). Among several tree species, Sapota (*Achras zapota*) is one that has been found to have very good economic value (Amutha, 2014). It is an evergreen, deep-rooted, drought-resistant and has good soil binding properties (Hernandez-Sanchez, Hernandez-Fuentes, Elorza-Martinez, Lopez-Herrera, & Lopez-Jimenez, 2009). Hence, it could be an excellent species to adopt for reclamation of ravine land of Western India. However, most afforestation studies on the degraded ravine lands have reported poor survival and growth of planted trees as a consequence of high soil erosion and low soil water holding capacity to sustain proper plant growth (Chaturvedi et al., 2014). Under such conditions, when high level of degradation, as in ravines, hampers tree growth and biomass production, proper SWCM needs to be integrated with afforestation (Kumar, Bhatnagar, Kakade, & Dobhal, 2020).

Worldwide, SWCM, including bench terracing and trenching, have been tested to restore ravine lands (Ali et al., 2016; Kurothe, Vishwakarma, et al., 2014). On one hand, several potential benefits such as checking runoff/soil erosion (García-Orenes et al., 2010), improving soil properties (Kaushal et al., 2016), promoting plant growth, enhancing biomass, and C assimilation and provisioning of commercial tree products (Chaturvedi et al., 2014) have been claimed with SWCM on different kinds of degraded lands. On the other hand, adverse effects such as an increase in the peak of soil erosion rates due to terracing are also not uncommon (Tarolli et al., 2015). Previous studies in this region have indicated significant effect of trenching densities (Ali et al., 2016; Kurothe et al., 2013), varying degree of slope (Kumar et al., 2014), vegetation cover (Rao, Mishra, Kurothe, Pande, & Kumar, 2015) and tillage operations (Kurothe et al., 2014) on runoff, soil loss, plant growth and productivity. Most of these studies point to an overall improvement in biological, hydrological and edaphic processes due to conservation measures. However, very limited studies exist on the impact of terraces compared to trenches on ravines, particularly in relation to tree growth characteristics, and runoff and soil erosion effects over longer period of time. Very little data exists on change in soil characteristics, biomass production trends, carbon stock and effects of drought periods. The effect of drought period may be an important criterion for adoption of tree species in such areas since ravines have poor water holding characteristics and adverse hydrological conditions. The role of SWCM in tree carbon assimilation and climate change mitigation, particularly drought tolerance deserves attention and this could assist those who plan rehabilitation measures (Jiao, Zou, Jia, & Wang, 2009), as presently less information is available regarding these aspects throughout the globe (Xu, Ma, & Fu, 2006).

Ravine areas of Western India are characterized by erratic and extreme rainfall events, low annual rainfall, frequent droughts, occasional flash flooding, high summer temperatures, severe erosion, and often, high anthropogenic pressure (Kumar et al., 2020). Therefore, in this study, we investigated the impact of SWCM on biology, hydrology and soil environment on degraded ravine land. We evaluated the performance of the Sapota tree, a sought-after species for degraded areas of Western India in conjunction with SWCM, bench terracing and trenching. Crop cultivation along with tree plantation was also evaluated as an option since seasonal crop yield can compensate for initial high costs of land modification as in bench terracing. The objectives of the study were to assess the impact of these strategies over longer periods after establishment, and evaluate periodic variations in plant growth and conservation effects (runoff, soil loss, and soil quality changes).

2 | MATERIALS AND METHODS

2.1 | Experimental site

The study was conducted on a semi-arid degraded ravine landscape located at Vasad, Anand, in Western India (latitude 22°16′N; longitude 72°58′E and altitude 25 m a.m.s.l). The regional climate is a tropical semiarid dry type with an average annual temperature of 25°C and rainfall of 871 mm yr^{-1} (Kurothe, Kumar, et al., 2014). During the experiment period, mean annual temperature was 25.4°C, with

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monthly mean temperatures ranging from 17.35°C in January to 33.51°C in May, and the mean annual precipitation was 983 mm (Figure 1). Around 90% of the annual rainfall is received during the monsoon period (July-Sept). The area experienced drought (475 mm) in 2015 and high rainfall (1,611 mm) in 2013, respectively. The native vegetation consisted of typical tropical dry deciduous and thorny species of *Acacia*, *Prosopis*, *Lantana* (shrub) and the annuals. The soil is classified as fine-loamy, mixed calcareous, hyperthermic, and falls under Fluventic Ustochrepts (associated with Inceptisols) (Soil Survey Staff, 2014). In general, soils are low in organic carbon (0.5%) and nitrogen (96 kg ha^{-1}), and the soil pH value is around 8.0.

2.2 | Experimental design and treatments

The study was initiated in 2010 to assess the potential of soil and water conservation measures (SWCM) in the restoration of degraded ravine lands. The experiment results were recorded for 6 years (2010–2015). Initially, the experimental site was cleared by removing scattered vegetation and thorny bushes to ensure plot uniformity and to design conservation measures for developing the production systems. Four plots (72 m length × 24 m width) parallel to the slope were established at the experimental site, each separated by an earthen bund (30 cm height). Two plots were converted to terraced land and divided into four sub-plots (terrace) to reduce slope length for soil-water conservation. The remaining two plots were maintained at a uniform slope (14%) for designing trench and control treatment. In one sloping plot, 96 trenches, sized 2 × 0.5 × 0.5 m were designed as SWCM, while the second sloping plot was maintained as a control. In this experiment, bench terracing and sloping designs were considered, keeping in mind the regional geography as bench terraces and slopes are common throughout the entire ravine landscape. Trenches were considered to observe their effectiveness in intercepting runoff' and water storage for plant growth. Trenching also needs very low soil disturbance compared to bench terracing. The four treatments (Figure 2), therefore, were (1) bench terracing + Sapota (Bt), (2) bench terracing + Sapota with crop cultivation (BtCr), (3) continuous slope + Sapota (Sl), and (4) continuous slope + Sapota with trenches (SlTr).

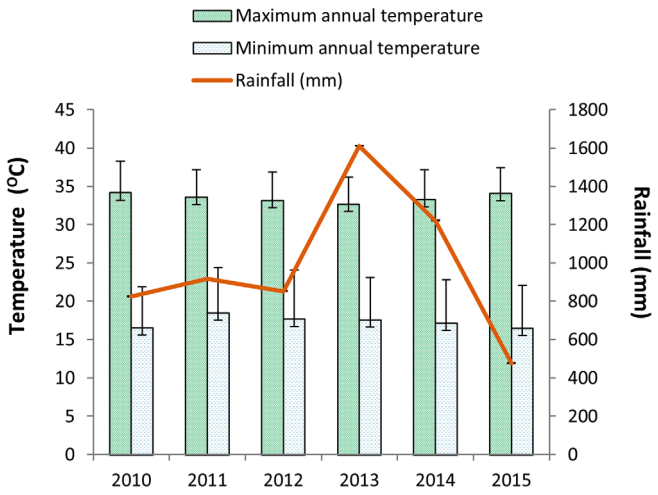


FIGURE 1 Annual rainfall, maximum annual temperature and minimum annual temperature recorded (mean ± SD, *n* = 12) at the experiment site (2010–2015) [Colour figure can be viewed at wileyonlinelibrary.com]

2.3 | Runoff and soil loss determination

A V-notch iron structure and stage level recorder was fixed at the outlet of each plot to measure soil loss and runoff, respectively. The runoff sediment samples were collected at the outlet of each plot during each rainfall event, from year 2010 to 2015 and later analyzed in the laboratory to determine sediment concentration and soil loss (Fang, Zhang & Cheng, 1999). The runoff was quantified from the runoff charts fixed in the stage level recorders using the procedure by Pathak, Wani, Singh, and Sudi (2004). The runoff coefficient was determined as the ratio of the amount of runoff to the amount of precipitation and expressed in term of percentage.

2.4 | Plantation and crop cultivation

In March 2010, 6-month-old seedlings of Sapota (*Achras zapota*) were planted in all of the four treatments in uniform number (27 in each treatment) at spacing of 8 m × 8 m. The recommended cultural practices were followed to maintain Sapota growth and productivity (Amutha, 2014). In BtCr treatment, castor (*Ricinus communis*) and cowpea (*Vigna unguiculata*) crops were grown along with Sapota to observe the effect of crop cultivation and tillage on soil loss and tree growth. The crops were grown annually in 1(castor):2(cowpea) proportion, and conventional tillage and standard recommended agronomic practices were followed.

2.5 | Measurement of trees

To evaluate Sapota growth and productivity, eight trees per treatment (two per replication) were randomly selected and measured quarterly every year (2010–2015). Basal diameter, total tree height, crown width, and fruit yield were determined in sampled trees using the procedures described by West (2009). The stem volume of trees was computed to estimate stem biomass using the methodology described by Tomar et al. (2015). The diameter (>2.5 cm) of all the branches was also recorded for each sampled tree.

2.6 | Biomass and carbon stock estimation

Destructive sampling of 25 branches of varying diameter (>2.5 cm) was carried out to determine dry biomass (leaf + branch biomass) in the Sapota tree as per the procedures by Poudel, Temesgen, and Gray (2015). The relationship between branch diameter (cm) and dry

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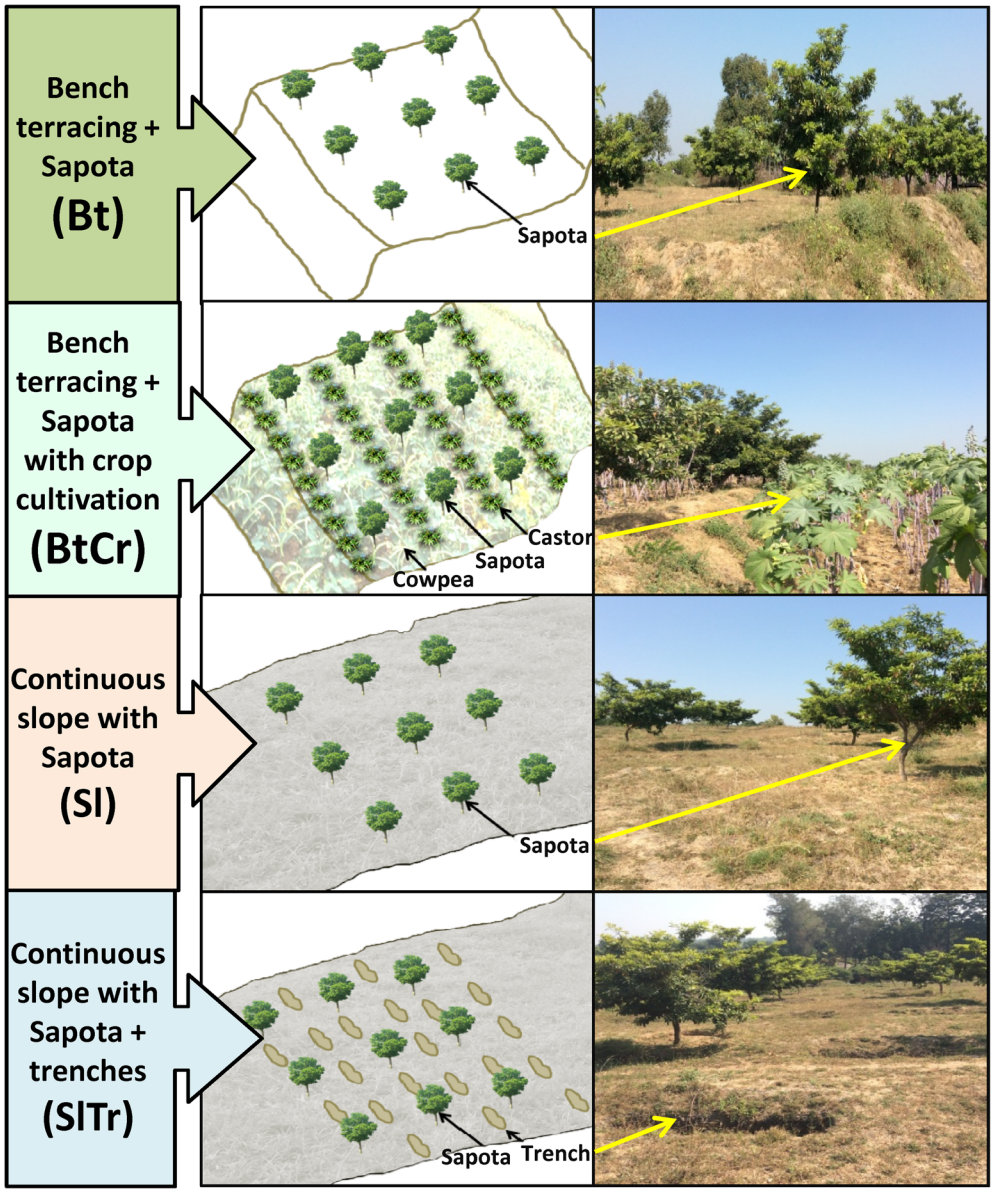


FIGURE 2 Treatment and layout of soil and water conservation measures. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; Sl = continuous slope + Sapota; SlTr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

biomass (kg) was established to compute the branch biomass (Table 1). The wood specific-gravity (0.93 g cm^{-3}) and root-shoot ratio (0.31) in Sapota were estimated as per methodology by Návar (2010). The above-ground biomass (stem wood, branch wood, and leaves), below-ground biomass and carbon stock were estimated in Sapota as per the procedure by Petersson et al. (2012).

2.7 | Soil analysis

In order to analyze the soil physic-chemical properties, a composite soil sample was made from four sub-samples which were collected at three different depths (0–15, 15–30 and 30–45 cm), for each treatment, every year (2010–2015). The soil pH and EC were determined using 1:2.5 soil–water suspension with a multimeter (Eutech Instruments, Singapore) (Jackson, 1973). The organic carbon was determined using wet digestion method (Walkley & Black, 1954). The

TABLE 1 Relationships and references used for estimation of biomass and carbon stock in Sapota

Sr. No	Relationships used
1.	Stem biomass = wood specific gravity (0.93 g cc^{-1}) × stem volume (Tomar et al., 2015)
2.	Branch (s) biomass =$0.1238e^{0.1090x}$ (x = branch diameter; e = exponential base) (Poudel et al., 2015)
3.	Above ground biomass = Stem biomass + Branch (s) biomass (Tomar et al., 2015)
4.	Root: shoot ratio (0.31) = Dry biomass of root/dry biomass of shoot (Návar, 2010)
5.	Below ground biomass = Above ground biomass × root: shoot ratio (IPCC, 2003)
6.	Total tree biomass = Above ground + below-ground biomass (Petersson et al., 2012)
7.	Carbon stock = Total tree biomass × 0.50 (Petersson et al., 2012)

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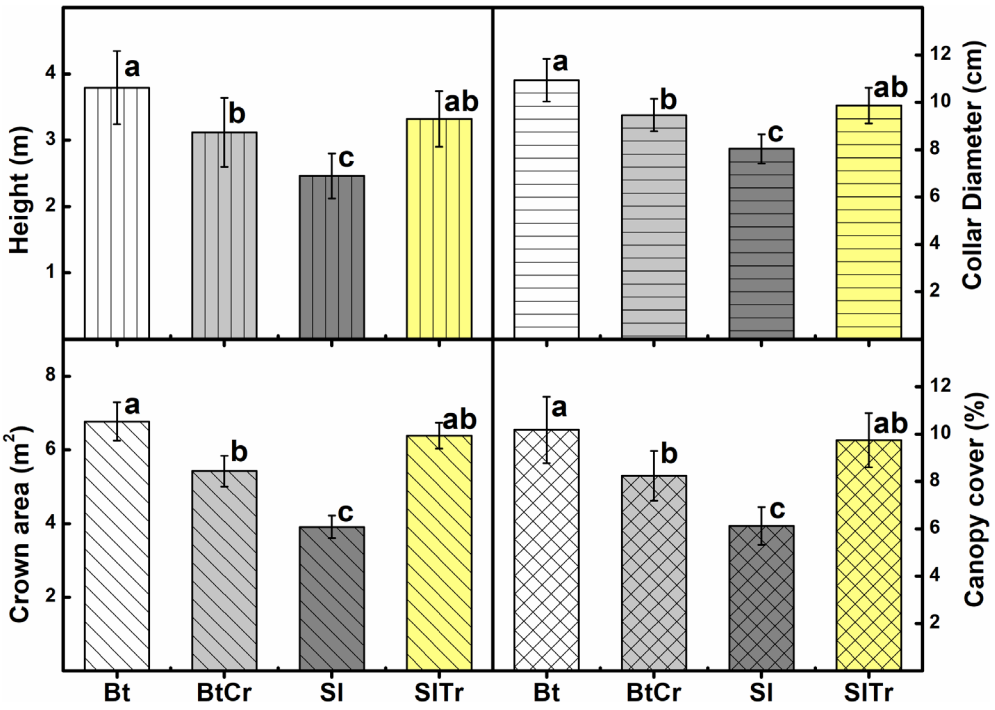


FIGURE 3 Effects of different treatments on plant growth parameters. The error bars denote ±1SE. Bars with same letters are not significantly different at $p \leq .05$. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; Sl = continuous slope + Sapota; SlTr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

available N, P, and K were determined by alkaline permanganate (Subbiah & Asija, 1956), $SnCl_2$ reduced phosphomolybdate (Jackson, 1973) and flame photometer methods, respectively.

2.8 | Statistical analysis

The Tukey's HSD (honestly significant difference) was performed in conjunction with one-way ANOVA with four replications in a randomized block design (quasi) to detect mean differences between various treatments for different parameters, except for the runoff and soil loss. The continuous slope + Sapota (Sl) treatment was considered as a control to compare with other treatments. The correlation coefficients among parameters were determined, and significance was analyzed using the *t*-test. The SPSS 16.0 software (IBM, USA) used for statistical analysis and the interactions among the treatments were tested at a 5% level of significance. The graphing was done using origin pro 8.5 (Origin Lab Corporation, Northampton).

3 | RESULTS

3.1 | Tree growth

Mean tree height and collar diameter in Bt (bench terracing + Sapota), BtCr (bench terracing + Sapota with crop cultivation), SlTr (continuous slope + Sapota with trenches) and Sl (continuous slope + Sapota) treatments ranged between 246–379 cm, and 8.0–10.9 cm, respectively (Figure 3). Height and diameter growth showed a significantly ($p < .05$) maximum increase in Bt, compared to SlTr, BtCr, and Sl. Tree height in SlTr, BtCr, and Sl treatments were recorded 12.4%, 17.7%, and 35% lower, respectively, compared to Bt. Likewise, compared to Bt, diameter growth was observed lesser in SlTr (9.8%), BtCr (13.4%) and Sl (26.4%). Overall, conservation measures such as bench terracing, bench terracing with crop cultivation and trenching improved height and diameter growth in Sapota, compared to the continuous slope. Moreover, crown area and canopy cover also portrayed more or less the same picture as height and diameter growth in all the treatments. These results showed that conservation measures enhanced the height and diameter growth of Sapota.

3.2 | Biomass and carbon stock

The soil and water conservation measures (SWCM) significantly ($p < 0.05$) affected biomass and carbon stock of Sapota (Figure 4). Bench terracing (Bt) improved biomass and carbon stock by 4.4 and 2.2 t ha^{-1}, respectively, compared to the continuous slope (Sl), while trenching (SlTr) had 2.9 and 1.5 t ha^{-1} increase, respectively. Similarly, cultivated bench terrace (BtCr) recorded 2.3 and 1.2 t ha^{-1} higher biomass and carbon stock, respectively, compared to the continuous slope (Sl). Overall, Biomass production and carbon stock was observed higher in bench terracing, followed by trenching, cultivated bench terrace and least on continuous slope, respectively. These results explained that conservation measures improved the biomass productions and carbon stock in Sapota.

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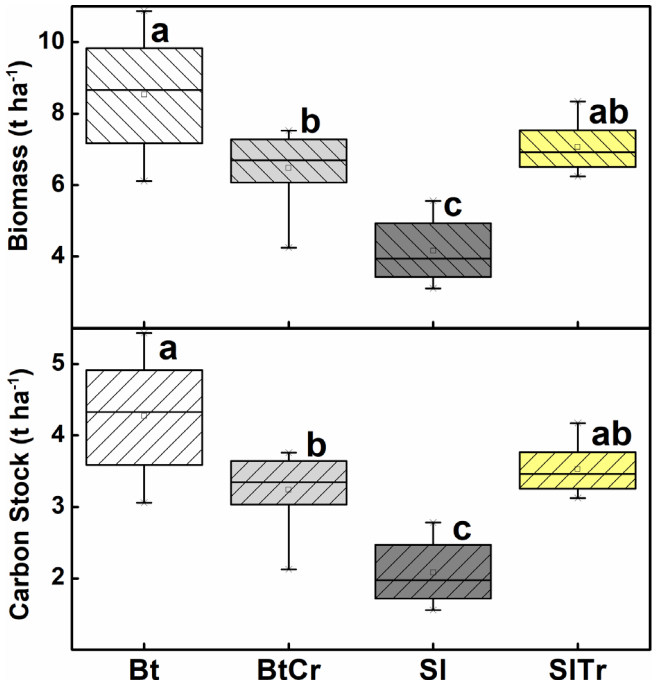


FIGURE 4 Biomass (t ha^{-1}) and carbon stock (t ha^{-1}) of Sapota under different treatments. The error bars denote ±1SE. Bars with same letters are not significantly different at $p \leq .05$. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; Sl = continuous slope + Sapota; SlTr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

3.3 | Runoff and soil loss

In our study, the mean annual runoff was recorded higher in SL followed by SlTr and BtCr compared to Bt, respectively (Figure 5). Similarly, the soil loss was recorded greater in BtCr followed by SL and SlTr compared to Bt. Quantitatively, bench terracing (Bt) and trenching (SlTr) reduced mean annual runoff by 1060 and 520 m^3 ha^{-1} yr^{-1}, respectively, while soil loss was reduced by 1.5 and 0.90 Mg ha^{-1} yr^{-1}, respectively, compared to the continuous slope (Sl). In contrast, bench terracing with crop cultivation (BtCr), produced 1.01 Mg ha^{-1} yr^{-1} higher soil loss and 850 m^3 ha^{-1} yr^{-1} lower runoff, compared to the continuous slope (Sl). The results showed that conservation measures (bench terracing and trenching) reduced the runoff and soil loss in degraded ravine lands.

3.4 | Soil physical and chemical properties

In this study, nitrogen (N), phosphorus (P), and potassium (K) in BtCr were recorded significantly ($p < .05$) higher compared to Bt, SlTr, and Sl, after 6 years (2015) (Table Data S1). However, a comparison among Bt, SlTr, and Sl revealed that organic C, N, P, and K were highest in Bt, followed by SlTr, and least in Sl. Change in soil physic-chemical properties from the year 2010–2015 showed the maximum change in soil N (+48.2%), P (+55.1%), and K (+27.5%) in bench terracing with crop cultivation (Figure 6). On the other hand, the least

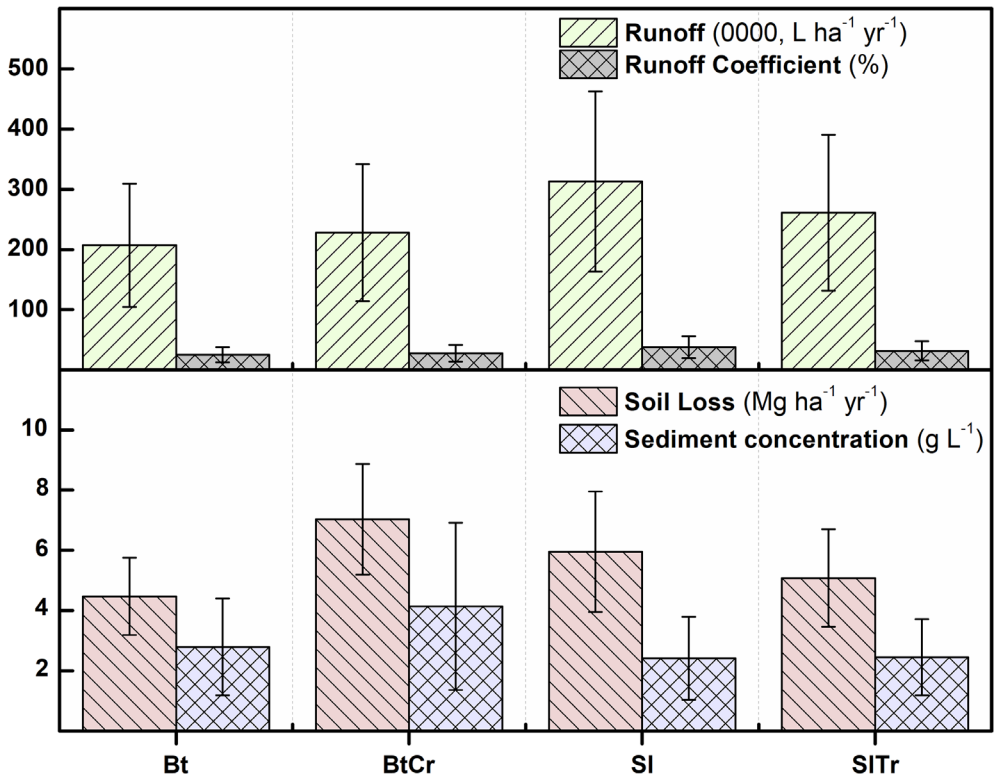


FIGURE 5 Average annual runoff, runoff coefficient, average annual soil loss and sediment concentration under different treatments during 6 years of study (2010–2015). The error bars denote ±1SD. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; Sl = continuous slope + Sapota; SlTr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

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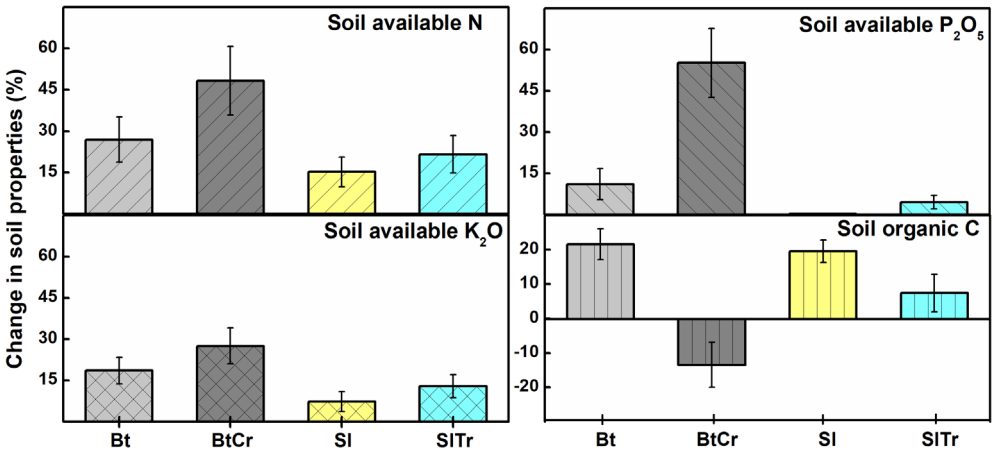


FIGURE 6 Changes in soil properties during the period of study (2010–2015). The error bars denote ±1SE. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; SI = continuous slope + Sapota; SITr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

change in soil N (15.1%), P (0.3%), and K (7.3%) was recorded in the continuous slope. Increase in organic carbon was recorded highest in bench terrcing (21.5%), and a decrease was recorded in bench terracing with crop cultivation (−13.4%), compared to the continuous slope. These results support the notion that conservation measures improved the soil properties in degraded ravine lands.

3.5 | Fruit yield during drought

The effects of SWCM on the fruit yield of Sapota are presented in Figure 7. During the study period, 2015 was a drought year and Bt had the highest fruit yield which was followed by BtCr, SITr, and SI, in decreasing order. In general, bench terracing (Bt), bench terracing with crop cultivation (BtCr) and trenching (SITr) had 604, 434, and 69 kg ha^{-1} more fruit production over continuous slope (SI) under drought conditions. These results demonstrated that conservation measures sustained fruit yield of Sapota during drought period.

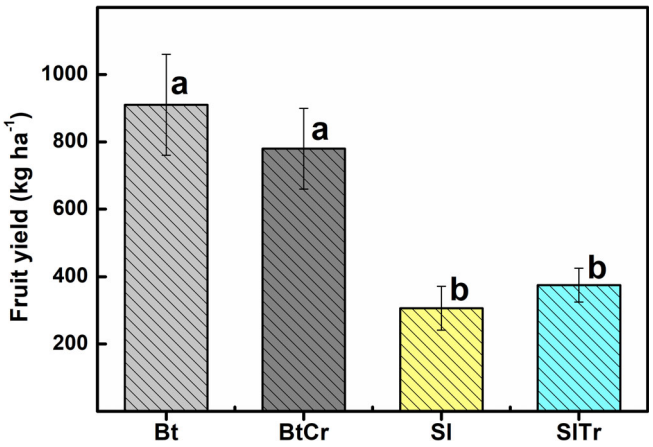


FIGURE 7 Fruit yield of Sapota under different treatments during drought period. The error bars denote ±1SE. Bars with same letters are not significantly different at $p \leq .05$. Bt = bench terracing + Sapota; BtCr = bench terracing + Sapota with crop cultivation; SI = continuous slope + Sapota; SITr = continuous slope + Sapota with trenches [Colour figure can be viewed at wileyonlinelibrary.com]

3.6 | Correlation between soils, plant and hydrological parameters

Significant ($p < .05$) correlation was observed between most of the parameters, except for the fruit yield (Table 2). Runoff appeared strongly correlated with most of the parameters except for height, diameter and K. Likewise, soil loss showed strong correlation with growth, biomass, organic carbon (C), and availability of N, P, and K. Overall, correlation study revealed that growth and biomass production is highly dependent on soil loss, organic C, soil N, soil P, and soil K.

4 | DISCUSSION

Ravine landscapes in Western India are severely affected by the soil erosion and rainfall aberrations which results in the reduced plant productivity and enhanced resource degradation. In this region, the number of rainfall events that actually contribute to runoff generation vary from 5 to 10 per annum (Kumar et al., 2014). High runoff is generated even after a moderate rainfall event that results in soil loss. Sometimes high rainfall events of more than 100 mm per day (once in a year or two) result in very high erosion that leads to formation of small gullies immediately, which turn into large ravines overtime if conservation measures are not implemented. Moreover, due to continuous soil loss and in the absence of any conservation measures, soils are unable to support any plant growth (Mehta et al., 2018). Therefore, any practice that provides opportunity for better nutrient availability and moisture retention would have profound effect on plant establishment and growth. Soil water conservation measures have demonstrated their potential in increasing rainwater infiltration into the soil that results in reduced runoff and soil loss as well as improved plant growth (Fernández, Fernández-Alonso, & Vega, 2020; Wang et al., 2016). Ravines have poor productivity because the soils are unable to support any plant growth. Poor water holding capacity

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TABLE 2 The correlation coefficients between the means of different parameters

	Height	Diameter	Biomass	Fruit yield	Runoff	Soil loss	OC	pH	EC	N	P	K
Height	1											
Diameter	0.93*	1										
Biomass	0.76*	0.85*	1									
Fruit yield	0.83	0.84	0.82	1								
Run off	-0.86	-0.87	-0.85*	--1.00	1							
Soil loss	-1.00*	-1.00*	-0.96*	-0.88	0.91*	1						
OC	0.97*	0.97*	0.96*	0.94	-0.96*	-0.99*	1					
pH	0.6*	0.61	0.58*	0.94	-0.92*	-0.67*	0.78*	1				
EC	0.94*	0.93*	0.94*	0.59	-0.63*	-0.9*	0.82*	0.28*	1			
N	0.98*	0.97*	0.98*	0.70	-0.74*	-0.95*	0.9*	0.41*	0.99*	1		
P	1.00*	1.00*	1.00*	0.81	-0.84*	-0.99*	0.96*	0.56*	0.95*	0.99*	1	
K	0.88*	0.89*	0.87*	1.00	-1.00	-0.92*	0.97*	0.9*	0.66*	0.76*	0.86*	1

Note: Colour coding: From yellow to green, Increasing more positive correlation. From yellow to red, Increasing more negative correlation. no colour = non-significant correlation (at $p < .05$).
*Significant correlation at $p < .05$

and moisture retention are the main reason but nutrient availability also plays a significant role. With erosion, most of the mineralized nutrients are lost and therefore plant growth is not supported. Any practice that makes significant differences to nutrient retention will result in enhanced productivity and favourable soil physical and chemical conditions for plant growth.

During 6 years (2010–2015) of experimentation, our results showed that the continuous slope produced the highest runoff and conservation measures like bench terracing and trenching reduced the runoff generation. Similar observation has been noted by Zhang et al. (2014, 2017) wherein terracing reduced the runoff and soil loss, and improved the vegetation growth. Crop cultivation on bench terrace (BtCr) only slightly increased runoff compared to non-cultivated bench terrace (Bt), but it substantially increased the soil loss. This indicated that even slight/moderate runoff events carried significant loads of soil when crop cultivation is combined with terracing (Figure 5). This was due to fact that conventional tillage operation resulted in loosening the soil surface. The initiation of the rainy season simultaneously resulted in high soil loss, particularly during the germination and seedling growth and before the establishment of good crop cover. Even after crop harvest (without digging roots), land is fallow that also results in high runoff and soil loss. In such conditions, rills are easily formed even on the terrace. Therefore, plant cover is perhaps one of the most important steps in rehabilitating ravine lands. Similar to our observation, Kurothe, Kumar, et al. (2014) also reported higher soil loss due to conventional tillage operations in a terraced ravine land. Our findings indicated that crop cultivation in between trees plantations is not preferable unless other stringent conservation measures like increased dike height or crop residue retention/mulching along with no-tillage is adopted. Since trenching also reduced soil loss significantly, compared to the continuous slope (Sl), it can also serve as a cost-effective measure under such circumstances. Also, staggered contour trenching in a triangular arrangement along the slope (calculated for the region based on Kurothe, Vishwakarma, et al., 2014) worked effectively for controlling the runoff on these soils. Distribution and density of trenches may greatly influence the effectiveness. Ali et al. (2016) also suggested that trench density has significant effect on runoff and soil loss in ravine lands. Therefore trench density and distribution need to be carefully evaluated before adoption. Yet, it is another parameter which can be varied and manipulated to achieve cost-effectiveness.

The significant improvement in soil organic matter in non-cultivated bench terrace and trenching resulted from the greater grass cover and its mixing in soil, and subsequently decomposition overtime time. On the other hand, a decrease in soil organic carbon in crop cultivation on bench terraces (BtCr) was noted, perhaps an effect of tillage and resulting expedited decomposition. Tillage exposes soil organic matter that expedites decomposition as well as increases its susceptibility to erosion during the rainfall events. In case of soil nutrients, there was increase in available nutrients stocks (N, P, and K) in terracing with crop cultivation (BtCr), probably because nutrients as fertilizers were added for cowpea and castor growth. The additional input of fertilizers is translated into increased nutrient content of soil. But interestingly, in non-cultivated terrace (Bt) and trench system (SlTr), even without any nutrient inputs, conservation of soil and water contributed increased nutrient content significantly. A similar effect has been noted by Jiao, Wang, and Li (1999), who observed the increased N and organic C content of soils on terraced land, compared to the slope. Similarly, SWCM have been noted to increase soil moisture content, nutrient retention and availability to plants (Singh, 2012; Zhuang et al., 2016). Moreover, in unfertile ravines, small amount of nutrient conservation through SWC along with synergistic effect of

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conserved soil moisture could have profound effect on plant growth. The above reasons are highly relevant during early years of the tree growth and establishment (i.e., shallow roots and erosion depleting top soil nutrients) under different conservation measures.

The improvement in soil properties, resulting from reduced soil and water loss under conservation measures, enhanced growth of Sapota tree. Before plantation, the study site was dominated by bushy vegetation, with scattered plant population and stunted growth, primarily because of low rainfall along with high soil erosion, runoff losses and infertile soil. In bench terracing and trenching, greater soil, nutrient and water conservation measure (SWCM) resulted in increased plant height and collar diameter, and improved the overall growth. In general, trench and bench terracing have been noted to create favorable physical, chemical and biological environment in soil that enhances the plant growth (Kebede, 2014; Raiesi & Salek-Gilani, 2020). Specifically, terracing without crop cultivation proved to be the best practice, while crop cultivation (on terraces) seems to have a little negative effect in term of growth. Moreover, if economics is considered then the crop cultivation may be more profitable. However, higher soil loss from this system may subsequently lead to the formation of rills and gullies. Supporting native or improved perennial grasses instead of crops may be a better option, especially since droughts are regular phenomena in this region and crop failure is common, while it little affect the grasses (Kumar et al., 2020). The crop cultivation option need to further evaluated in terms of economics and its longterm sustainability.

Overall the favorable changes in ravine ecosystems, due to conservation measures and tree plantations have longterm implications for climate change mitigation and adaptation via enhanced carbon sequestration and climate resilience (Franzluebbers & Doraiswamy, 2007; Pande et al., 2018). Our finding approved that carbon stocks were significantly increased in the conservation treatments of bench terracing and trenching. The conditions in ravines are also highly fluctuating in terms of rainfall, and therefore survival of plants is difficult in the absence of any conservation measure. Droughts are the common phenomena in the areas with the ravines, particularly in Western India. Any conservation measure which has capacity to make plants withstand drought period has the potential to be accepted by the community. In fact this could be the most important criterion in the ravine lands of Western India which have frequent drought years. The superiority of the bench terracing over trenching and continuous slope showed its potential to withstand droughts and extremes climates in term of fruit yield. However, performance of these measures needs to be further investigated for prolonged drought conditions. Conservation measures have been noted to maintain growth and productivity of plants during drought conditions, earlier (Zhu & Zhao, 2003). Ravine lands are an integral part of the livelihood of the marginal landholders, mostly farmers, and the landholders depend for their livelihood on these lands. Slight improvement in productivity of these lands can improve the livelihood of these people. In long term, these improvements can also serve the purposes of climate change mitigation and carbon sequestration which is an urgent goal, worldwide.

5 | CONCLUSIONS

The results of this study indicated that bench terracing is the best conservation measure for the rehabilitation of ravine lands of Western India, and trenching is the second-best option. The choice between the bench terracing and trenching is a matter of options available in term of financial implications and landscape accessibility to heavy earth moving machinery. Bench terracing and trenching brought in significant decrease in runoff (34 and 16%, respectively) and soil erosion (25 and 15%, respectively), and enhanced tree growth, biomass and carbon stock. Crop cultivation in between tree plantations may induce significant soil loss (18% higher) even though runoff is not significantly affected. Crop cultivation may be an important means to compensate for cost of land modification yet, it needs to be carefully evaluated. Conservation measure like residue retention/mulching and raised dikes need to be implemented when crop cultivation is adopted. In case bench terracing itself is not feasible then trenching in between the tree rows and oriented perpendicular to slope, would serve significant purpose for rehabilitation of highly degraded ravines of Western India. These conservation measures could also be important strategies for climate change mitigation and adaptation for such highly degraded and marginally productive regions of the world where slight improvements can have substantial impact on the environment as well as society.

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CONFLICT OF INTEREST

The authors have no conflict of interest.

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REFERENCES

Ali, S., Sethy, B. K., Singh, R. K., Parandiyal, A. K., & Kumar, A. (2016). Quantification of hydrologic response of staggered contour trenching for horti-pastoral land use system in small ravine watersheds: A paired watershed approach. *Land Degradation & Development*, *20*, 1237–1252. https://doi.org/10.1002/ldr.2517

Amutha, D. (2014). An analysis of Sapota fruit cultivation in Tuticorin District. *International Journal of Economics & Management Sciences*, *3*, 203. https://doi.org/10.4172/2162-6359.1000203

Biratu, A. A., & Asmamaw, D. K. (2016). Farmers' perception of soil erosion and participation in soil and water conservation activities in the Gusha Temela watershed, Arsi, Ethiopia. *International Journal of River Basin Management*, *14*(3), 329–336. https://doi.org/10.1080/15715124.2016.1167063

1091145x, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ldr.3691 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [17/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Brauch, H. G., & Spring, U. O. (2009). *Securitizing the ground grounding security UNCCD issue paper no. 2* (pp. 1–52). Bonn, Germany: Secretariat of the United Nations Convention to Combat Desertification. Retrieved from http://catalogue.unccd.int/843_dldd_eng.pdf

Cai, X., Zhang, X., & Wang, D. (2011). Land availability for biofuel production. *Environmental Science & Technology*, *45*(1), 334–339. https://doi.org/10.1021/es103338e

Chaturvedi, O. P., Kaushal, R., Tomar, J. M. S., Prandiyal, A. K., & Panwar, P. (2014). Agroforestry for wasteland rehabilitation: Mined, ravine, and degraded watershed areas. In J. C. Dagar, A. Singh, & A. Arunachalam (Eds.), *Agroforestry systems in India: Livelihood security & ecosystem services Advances in Agroforestry* (Vol. 10, pp. 233–271). New Delhi: Springer. https://doi.org/10.1007/978-81-322-1662-9_8

de MoraesSá, J. C., Séguy, L., Tivet, F., Lal, R., Bouzinac, S., Borszowskei, P. R., … Friedrich, T. (2015). Carbon depletion by plowing and its restoration by no-till cropping systems in oxisols of subtropical and tropical agro-ecoregions in Brazil. *Land Degradation & Development*, *26*, 531–543. https://doi.org/10.1002/ldr.2218

Evans, D. M., Zipper, C. E., Burger, J. A., Strahm, B. D., & Villamagna, A. M. (2013). Reforestation practice for enhancement of ecosystem services on a compacted surface mine: Path toward ecosystem recovery. *Ecological Engineering*, *51*, 16–23. https://doi.org/10.1016/j.ecoleng.2012.12.065

Fang, Y., Zhang, H., & Cheng, Y. (1999). Progress of sediment concentration measurement. *Engineering Journal of Wuhan University*, *3*, 55–57.

Fernández, C., Fernández-Alonso, J. M., & Vega, J. A. (2020). Exploring the effect of hydrological connectivity and soil burn severity on sediment yield after wildfire and mulching. *Land Degradation & Development*, 1–11. https://doi.org/10.1002/ldr.3539

Franzluebbers, A. J., & Doraiswamy, P. C. (2007). Carbon sequestration and land degradation. In M. V. K. Sivakumar & N. Ndiang'ui (Eds.), *Climate and land degradation. Environmental science and engineering (environmental science)*. Berlin, Heidelberg: Springer.

García-Orenes, F., Guerrero, C., Roldán, A., Mataix-Solera, J., Cerdà, A., Campoy, M., … Caravaca, F. (2010). Soil microbial biomass and activity under different agricultural management systems in a semiarid Mediterranean agroecosystem. *Soil and Tillage Research*, *109*, 110–115. https://doi.org/10.1016/j.still.2010.05.005

Hernandez-Sanchez, M., Hernandez-Fuentes, A. D., Elorza-Martinez, P., Lopez-Herrera, M., & Lopez-Jimenez, M. A. (2009). Fruit characteristics of star apple (*Chrysophyllum cainito* L.) at Veracruz State, Mexico. *Revista UDO Agrícola*, *9*(1), 70–73. https://tspace.library.utoronto.ca/handle/1807/45439

Hu, Y. L., Niu, Z. X., Zeng, D. H., & Wang, C. Y. (2015). Soil amendment improves tree growth and soil carbon and nitrogen pools in Mongolian pine plantations on post-mining land in Northeast China. *Land Degradation & Development*, *26*, 807–812. https://doi.org/10.1002/ldr.2386

IPCC. (2003). *Good practice guidance for land use, land-use change and forestry*. Hayama, Japan: The Intergovernmental Panel on Climate (IPCC).

Jackson, M. L. (1973). *Soil chemical analysis* (pp. 20–30). New Delhi, India: Prentice Hall of India Pvt. Ltd.

Jiao, J., Zou, H., Jia, Y., & Wang, N. (2009). Research progress on the effects of soil erosion on vegetation. *Acta Ecologica Sinica*, *29*, 85–91. https://doi.org/10.1016/j.chnaes.2009.05.001

Jiao, J. Y., Wang, W. Z., & Li, J. (1999). Analysis on soil and water conservation benefit of level terrace under different rainfall condition in loess hilly region. *Journal of Soil Erosion Soil Water Conservation*, *5*, 59–63. https://www.semanticscholar.org/paper/Analysis-on-Soil-and-Water-Conservation-Benefit-of-Ju/69d0a93491944a75f836f225c326429d29528268

Kaushal, R., Verma, A., Mehta, H., Mandal, D., Tomar, J. M. S., Jana, C., … Chaturvedi, O. P. (2016). Soil quality under *Grewia optiva* based agroforestry systems in Western sub-Himalaya. *Range Management and Agroforestry*, *37*(1), 50–55. http://www.indianjournals.com/ijor.aspx?target=ijor:rma&volume=37&issue=1&article=008

Kebede, W. (2014). Effect of soil and water conservation measures and challenges for its adoption: Ethiopia in focus. *Journal of Environmental Science and Technology*, *7*, 185–199. https://doi.org/10.3923/jest.2014.185.199

Keesstra, S. D. (2007). Impact of natural reforestation on floodplain sedimentation in the Dragonja basin, SW Slovenia. *Earth Surface Processes and Landforms*, *32*, 49–65. https://doi.org/10.1002/esp.1360

Kumar, G., Kurothe, R. S., Brajendra, P., Vishwakarma, A. K., Rao, B. K., & Pande, V. C. (2014). Effect of farmyard manure and fertilizer application on crop yield, runoff and soil erosion and soil organic carbon under rainfed pearl millet (*Pennisetum glaucum*). *Indian Journal of Agricultural Sciences*, *84*(7), 816–823.

Kumar, R., Bhatnagar, P. R., Kakade, V., & Dobhal, S. (2020). Tree plantation and soil water conservation enhances climate resilience and carbon sequestration of agro ecosystem in semi-arid degraded ravine lands. *Agricultural and Forest Meteorology*, *282-283*, 107857. https://doi.org/10.1016/j.agrformet.2019.107857

Kurothe, R. S., Kumar, G., Singh, R., Singh, H. B., Tiwari, S. P., Vishwakarma, A. K., … Pande, V. C. (2014). Effect of tillage and cropping systems on runoff, soil loss and crop yields under semiarid rainfed agriculture in India. *Soil and Tillage Research*, *140*, 126–134. https://doi.org/10.1016/j.still.2014.03.005

Kurothe, R. S., Singh, H. B., Tiwari, S. P., Pande, V. C., Bagdi, G. L., Sena, D. R., … Kumar, G. (2013). *Fifty years of research in soil and water conservation* (pp. 10–100). Research Centre, Vasad, India: CSWCRTI.

Kurothe, R. S., Vishwakarma, A. K., Sena, D. R., Kumar, G., Rao, B. K., & Pande, V. C. (2014). Decision support system for contour trenching. *Indian Journal of Soil Conservation*, *42*(2), 143–153. https://www.indianjournals.com/ijor.aspx?target=ijor:ijsc&volume=42&issue=2&article=004

Lal, R. (2001). Potential of desertification control to sequester carbon and mitigate the greenhouse effect. *Climate Change*, *51*, 35–72. https://doi.org/10.1023/A: 1017529816140

Lal, R. (2016). Soil carbon sequestration potential of degraded lands. In *Proceedings of the Global Ravine Conference on Managing Ravines for Food and Livelihood Security* (pp. 15–22). Madhya Pradesh (India): Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya.

Mehta, H., Kumar, R., Dar, M. A., Juyal, G. P., Patra, S., Dobhal, S., … Mishra, P. K. (2018). Effect of geojute technique on density, diversity and carbon stock of plant species in landslide site of North West Himalaya. *Journal of Mountain Science*, *15*, 1961–1971. https://link.springer.com/article/10.1007/s11629-017-4768-2

Millennium Ecosystem Assessment. (2005). *Ecosystems and human well-being: Desertification synthesis* (1–26). Washington, DC: World Resources Institute. Retrieved from http://www.millennium.assessment.org/documents/document.355.aspx.pdf

Minnemeyer, S., Laestadius, L., & Sizer, N. (2011). *A world of opportunity*. Washington, DC: World Resources Institute. Retrieved from www.wri.org/restoringforests

Munoz-Rojas, M., Jordan, A., Zavala, L. M., Dela, R. D., Abd-Elmabod, S. K., & Anaya-Romero, M. (2012). Organic carbon stocks in Mediterranean soil types under different land uses (southern Spain). *Solid Earth*, *3*, 375–386. https://doi.org/10.5194/se-3-375-2012

Návar, J. (2010). Measurement and assessment methods of forest aboveground biomass: A literature review and the challenges ahead. In M. Momba & F. Bux (Eds.), *Biomass* (p. 202). Croatia: Sciyo.

Navar, J., & Synnott, T. J. (2000). Soil infiltration and land use in Linares, N.L., Mexico. *Terra Latinoamericana*, *18*, 255–262.

Pande, V. C., Bhatnagar, P. R., Kumar, R., Dhakshanamoorthy, D., & Kumar, G. (2018). Mahi ravine ecosystem services- farmers' perception, services and incentives for management. *Technical Bulletin, ICAR-Indian Institute of Soil and Water Conservation, Research Center, Vasad, Anand, Gujarat, India*, 1–40. https://doi.org/10.13140/RG.2.2.14768.00000

1091145x, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ldr.3691 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [17/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Parandiyal, A. K., Samara, J. S., Singh, K. D., Singh, R., & Rathore, B. L. (2000). Floristic diversity of Chambal ravines under varying levels of protection. *Indian Journal of Soil Conservation*, *28*, 160–166.

Pathak, P., Wani, S. P., Singh, P., & Sudi, R. (2004). Sediment flow behaviour from small agricultural watersheds. *Agricultural Water Management*, *67*(2), 105–117. http://dx.doi.org/10.1016/j.agwat.2004.02.005

Perring, M. P., Standish, R. J., Price, J. N., Craig, M. D., Erickson, T. E., Ruthrof, K. X., … Hobbs, R. J. (2015). Advances in restoration ecology: Rising to the challenges of the coming decades. *Ecosphere*, *6*, 1–25. https://doi.org/10.1890/ES15-00121.1

Petersson, H., Holm, S., Stahl, G., Alger, D., Fridman, J., Lehtonen, A., … Makipaa, R. (2012). Individual tree biomass equations or biomass expansion factors for assessment of carbon stock changes in living biomass – A comparative study. *Forest Ecology and Management*, *270*, 78–84. https://doi.org/10.1016/j.foreco.2012.01.004

Poudel, K. P., Temesgen, H., & Gray, A. N. (2015). Evaluation of sampling strategies to estimate crown biomass. *Forest Ecosystem*, *2*, 1. https://doi.org/10.1186/s40663-014-0025-0

Prosdocimi, M., Jordán, A., Tarolli, P., Keesstra, S., Novara, A., & Cerdà, A. (2016). The immediate effectiveness of barley straw mulch in reducing soil erodibility and surface runoff generation in Mediterranean vineyards. *Science of the Total Environment*, *547*, 323–330. https://doi.org/10.1016/j.scitotenv.2015.12.076

Raiesi, F., & Salek-Gilani, S. (2020). Development of a soil quality index for characterizing effects of land use changes on degradation and ecological restoration of rangeland soils in a semiarid ecosystem. *Land Degradation & Development*, 1–12. https://doi.org/10.1002/ldr.3553

Rao, B. K., Gaur, M. L., Kumar, G., Kurothe, R. S., & Tiwari, S. P. (2013). Morphological characteristics and alterations in cross sections of different order streams of Mahi ravines. *Indian Journal of Soil conservation*, *41*(1), 20–24.

Rao, B. K., Mishra, P. K., Kurothe, R. S., Pande, V. C., & Kumar, G. (2015). Effectiveness of *Dichanthium annulatum* in watercourses for reducing sediment delivery from agricultural watersheds. *Clean Soil, Air and Water*, *43*(5), 710–716. https://doi.org/10.1002/clen.201400265

Roa-Fuentes, L. L., Martínez-Garza, C., Etchevers, J., & Campo, J. (2015). Recovery of soil C and N in a tropical pasture: Passive and active restoration. *Land Degradation & Development*, *26*, 201–210. https://doi.org/10.1002/ldr.2197

Safriel, U., Adeel, Z., Niemeijer, D., Puigdefabregas, J., White, R., Lal, R., … Archer, E. (2005). Dryland systems. In N. Ash, R. Hassan, & R. Scholes (Eds.), *Ecosystems and human well-being: Current state and trends Findings of the condition and trends working group* (pp. 623–662). Washington, DC: Island Press. Retrieved from https://www.millenniumassessment.org/documents/document.356.aspx.pdf

Shrivastava, P., & Kumar, R. (2015). Soil salinity: A serious environmental issue and plant growth promoting bacteria as one of the tools for its alleviation. *Saudi Journal of Biological Sciences*, *22*(2), 123–131. http://dx.doi.org/10.1016/j.sjbs.2014.12.001

Sikka, A. K., Mishra, P. K., Singh, R. K., Rao, B. K., & Islam, A. (2016). Management of ravines for food, livelihood, and environment security. In *Proceedings of the Global ravine conference on managing ravines for food and livelihood security* (pp. 1–14). Madhya Pradesh (India): Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya.

Singh, G. (2012). Enhancing growth and biomass production of plantation and associated vegetation through rainwater harvesting in degraded hills in southern Rajasthan, India. *New Forests*, *43*, 349–364. https://doi.org/10.1007/s11056-011-9285-5

Soil Survey Staff. (2014). *Keys to soil taxonomy* (12th ed.). Washington, DC: USDA-NRCS. Retrieved from https://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/survey/class/taxonomy/?cid=nrcs142p2_053580

Subbiah, B. V., & Asija, G. L. (1956). A rapid procedure for the estimation of available nitrogen in soils. *Current Science*, *25*, 259–260.

Tarolli, P., Sofia, G., Calligaro, S., Prosdocimi, M., Preti, F., & Dalla, F. G. (2015). Vineyards in terraced landscapes: New opportunities from lidar data. *Land Degradation & Development*, *26*(1), 92–102. https://doi.org/10.1002/ldr.2311

Tomar, J. M. S., Rathore, A. C., Kaushal, R., Kumar, R., Jayaprakash, J., & Chaturvedi, O. P. (2015). *Agroforestry: A research manual* (p. 120). Dehradun, India: M/s Bishen Singh Mahendra Pal Singh.

Walkley, A. J., & Black, I. A. (1954). Estimation of soil organic carbon by chronic acid titration method. *Soil Science*, *37*, 29–38.

Wang, J., Watts, D. B., Meng, Q., Zhang, Q., Wu, F., & Torbert, H. A. (2016). Soil water infiltration impacted by maize (*Zea mays* L.) growth on sloping agricultural land of the Loess Plateau. *Journal of Soil and Water Conservation*, *71*(4), 301–309. https://doi.org/10.2489/jswc.71.4.301

Wang, T., Xue, X., Zhou, L., & Guo, J. (2015). Combating aeolian desertification in Northern China. *Land Degradation & Development*, *26*, 118–132. https://doi.org/10.1002/ldr.2190

West, P. W. (2009). *Tree and forest measurement* (2nd ed., pp. 1–192). Berlin: Springer. https://doi.org/10.1007/978-3-540-95966-3

Xu, X. L., Ma, K. M., & Fu, B. J. (2006). A research review of the relationship between vegetation and soil loss. *Acta Ecologica Sinica*, *26*(9), 3137–3143.

Yan, X., & Cai, Y. L. (2015). Multi-scale anthropogenic driving forces of karst rocky desertification in Southwest China. *Land Degradation & Development*, *26*, 193–200. https://doi.org/10.1002/ldr.2209

Zhang, F., Xing, Z., Rees, H. W., Dong, Y., Li, S., & Meng, F. (2014). Assessment of effects of two runoff control engineering practices on soil water and plant growth for afforestation in a semi-arid area after 10 years. *Ecological Engineering*, *64*, 430442. https://doi.org/10.1016/j.ecoleng.2013.12.024

Zhang, F., Xing, Z. S., Zhao, C. Y., Deng, J. L., Yang, B., Tian, Q., … Badreldin, N. (2017). Characterizing long-term soil and water erosion and their interactions with various conservation practices in the semi-arid Zulihe basin, Dingxi, Gansu, China. *Ecological Engineering*, *106*, 458–470. https://doi.org/10.1016/j.still.2018.10.024

Zhu, Z. J., & Zhao, K. R. (2003). Study on afforestation technology through runoff regulation in arid and semi-arid regions. *Soil Water Conservation China*, *9*, 26–28.

Zhuang, S. Li, Q., Lv, P., & Gui, R. (2016). Effect of liming on soil nitrogen loss in runoff from a cultivated bamboo stand. *Journal of Soil and Water Conservation*, *71*(4), 356–363. https://doi.org/10.2489/jswc.71.4.356

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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